

Project Specification

CPTS 484: Software Requirements

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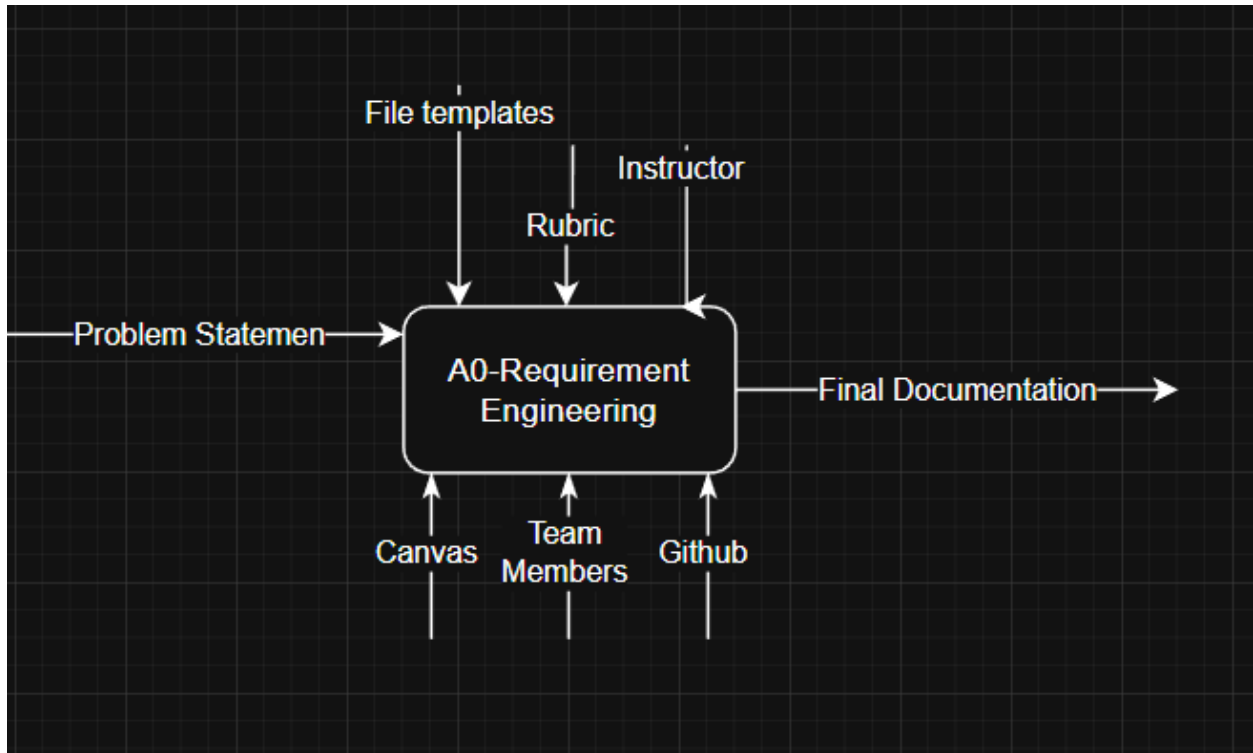
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Tools:

All graphs were made using draw.io due to the ease of use and accessibility of any components necessary for these diagrams. Additionally, the website is one that was used commonly in previous sections to produce class diagrams and other graphs which in turn helped provide experience and familiarity with this system. For these reasons draw.io was used for the graphs in this project specification.

A-0 Diagram:



Inputs:

Problem Statement:

- We as a group were tasked with researching the requirements of a theoretical app to assist the visibly impaired traverse through buildings

Control:

- File Templates provided allowed us to follow a guideline in our production of our documents. These templates could also be considered both inputs and control variables since these documents are both provided and control what we output.
- Rubric is the grading scale that our documents will be based on. This document details the expectations of quality and features our documents must include.

- Instructor provides us with guidance in every weekly module. The instructor helps guide the development of documents for our final submission and will assist in determining the quality and substance of our work

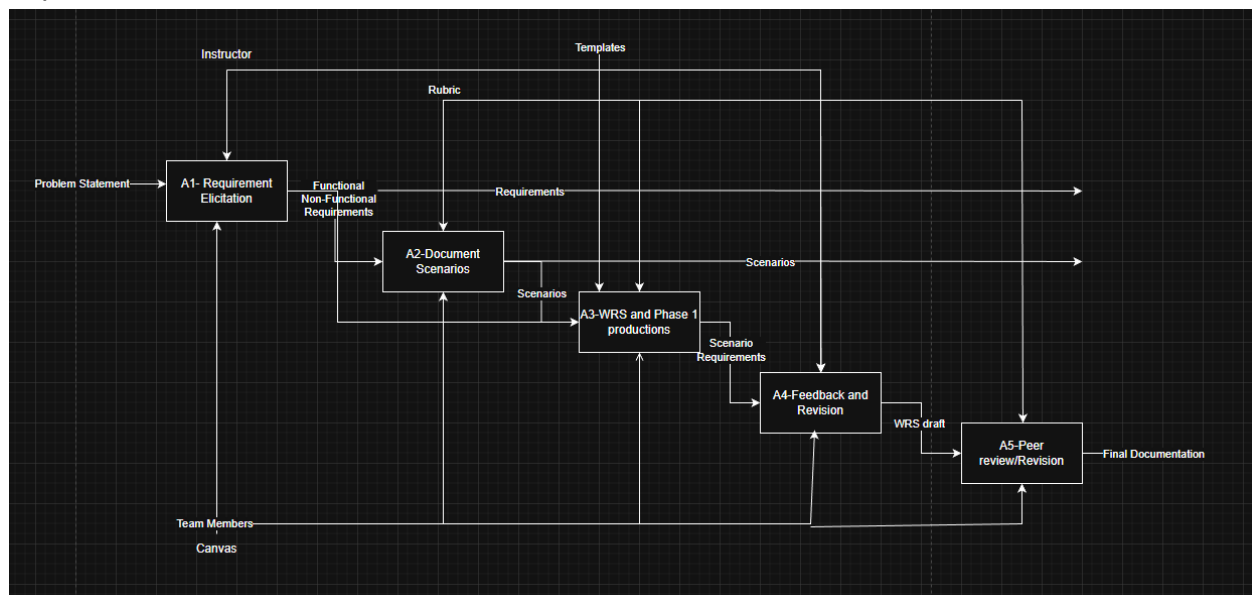
Mechanisms:

- GitHub is our primary storage platform where most documents and development will occur.
- Canvas is one of the primary deployment websites where we would submit our work for quality assurance from the instructor and revise our work to meet proper guidelines and rubric.
- Team members together will help develop our documents. Together we collaborated with one another and researched our requirements for our final products. Additionally, we provided a variety of perspectives and peer review to help create a quality product.

Output:

- Final documentation will summarize our work throughout the semester. These documents will include a WRS which will display our findings that we found during our development of our scenarios. This will include the various functional and non-functional requirements as well as our solutions for these systems and more detailed or developed responses to our scenarios introduced in phase 1.

Expanded IEDO:



A1 Requirement Elicitation:

In this activity we were required to research the necessary components and ideas that would make up our project. We divided the various requirements gathering activities among our group and assisted in peer reviewing them. We mostly brainstormed our ideas as we did not interview our targeted audience or users and went with the assumptions of the problem statement. We would further refine our ideas into more well-defined terms. Doing so allowed us to clearly communicate what our goals are for our system. We as a team will help define our system requirements efficiently allowing us to work effectively when we continue to document more complex ideas such as scenarios.

- Input: The problem statement provided the necessary context to begin
- Control: Instructor provides our environment and context to establish our initial ideas for our project
- Output: Developed ideas on how our system should behave and function.
- Mechanisms: Team members will help review our ideas and provide early feedback.

A2 Document Scenarios:

With our defined requirements for the project, we can then build more complex ideas such as scenarios for our project. Together we thought up a variety of context and situations our system will handle and establish how our program will handle these environments. Additionally for our group we had to prioritize and differentiate the scenarios with both feasibility and our ability to implement the features necessary for the scenarios. These responses were guided by our previous requirements to remain in line with our goals. The final output will help provide ideas on how we should implement our system around these scenarios and what our prototype goals will be.

- Input: Requirements that assist in the development of our scenarios
- Control: We followed a rubric that detailed how our scenarios will be explained and evaluated according to our class standards
- Output: Various situations our system should be able to handle and help layout the foundation for our features.
- Mechanisms: We would submit our work through canvas and team members had collaborated to brainstorm the various scenarios our project will be placed in.

A3 WRS and Phase 1:

The WRS document would help summarize our initial half of our semester where we would put what we have learned and produced into a single, well structured, and detailed file. We clearly displayed what we had produced for our project which included all our previous work of mostly requirements and scenarios into a professional document.

- Input: Requirements and scenarios from previous sections
- Control: We as a group followed a template outlining what the expectations on how the document should be made and constructed. What content should and should not be included from our previous work.
- Output: A WRS of our initial systems for phase 2 of our documentation.
- Mechanisms: We used Canvas as our main location for submitting documentation and receiving feedback or responses from instructors.

A1 Feedback and Revision:

Given our initial draft we were provided with various feedback and ideas that we should pursue from our instructor in Phase 2 of our documentation. These revisions would include ideas such as KAOS models, vision documents, and IEDO diagrams to include for our next phase of documentation. We would then implement general systems of our program as we began to develop the project. Doing so allowed us to gain further insight into what requirements and objectives we can realistically accomplish with our project as development continues. This would help mold and shape our ideas for the final revision.

- Input: Our WRS and other phase 1 documents.
- Control: The instructor provided feedback and guidelines that we should follow for phase 2 documents
- Output: A revised WRS which includes new sections for regrading and new analysis.
- Mechanisms: We would receive our grades via canvas and comments would help guide us on what we should improve on.

A5 Peer review/revision:

Our final revision before final submission of our document. We as a group would go over all the documents so everything should be for final submissions. As a group we would ensure that documents fit our vision and our helps ensure we are satisfied with our final product. This allows all members to provide input into each document and ensure there is a consensus on what we are submitting so there is no future confusion on our documents.

- Input: All previous documents from phase 1 and that will be revised as well as new documents and KAOS graphs.
- Control: Final rubric and instructor providing guidance on what we should be focusing on in our documents.
- Output: Our final complete product of our semester of requirement engineering.
- Mechanisms: Should include GitHub as we submitted our code alongside our documents on canvas.